

Supplementary Material

A Benchmark of Five Machine-Learning Architectures for Daily Trading-Signal Classification on US Tech Equities

MICAI 2026 — Lecture Notes in Artificial Intelligence — Chapter 171490174

This document contains the material that the paper refers to as “supplementary material”. Each section corresponds to one reference in the paper. The results in Sections S2–S4 come from the benchmark reported in Section 7 of the paper (120 runs, test year 2025); for the sequence models (LSTM, CNN 1D and CNN-LSTM) the reported run is the better of the two look-backs (20 or 60 days) by test F1-macro, as in the paper. They reproduce Tables 4, 5 and 6 of the paper.

S1 Full list of the 61 input features (paper §3.3)

All features are computed per ticker from daily OHLCV data adjusted for splits and dividends (O, H, L, C, V) and, for the last family, from SPY and the CBOE VIX, using only information available at the close of day t . Column names are those used in the code.

Table S1: Input features, grouped by family (count in parentheses).

#	Column	Definition
<i>Log returns</i> (5)		
1	ret_1d	$\ln(C_t/C_{t-1})$
2	ret_2d	$\ln(C_t/C_{t-2})$
3	ret_3d	$\ln(C_t/C_{t-3})$
4	ret_5d	$\ln(C_t/C_{t-5})$
5	ret_10d	$\ln(C_t/C_{t-10})$
<i>Momentum</i> (4)		
6	mom_5d	$C_t/C_{t-4} - 1$
7	mom_10d	$C_t/C_{t-9} - 1$
8	mom_20d	$C_t/C_{t-19} - 1$
9	mom_60d	$C_t/C_{t-59} - 1$
<i>Moving averages</i> (9)		
10	dist_ma10	$C_t/\text{MA}_{10} - 1$, with MA_k the k -day simple moving average of C
11	dist_ma20	$C_t/\text{MA}_{20} - 1$
12	dist_ma30	$C_t/\text{MA}_{30} - 1$
13	dist_ma50	$C_t/\text{MA}_{50} - 1$
14	dist_ma200	$C_t/\text{MA}_{200} - 1$
15	cruce_ma10_ma50	$\text{MA}_{10}/\text{MA}_{50} - 1$ (crossover)
16	cruce_ma20_ma50	$\text{MA}_{20}/\text{MA}_{50} - 1$ (crossover)
17	cruce_ma50_ma200	$\text{MA}_{50}/\text{MA}_{200} - 1$ (crossover)
18	pendiente_ma20	$(\text{MA}_{20,t} - \text{MA}_{20,t-5})/\text{MA}_{20,t-5}$ (slope)
<i>Volatility</i> (8)		
19	atr_14	ATR(14): 14-day mean of $\text{TR}_t = \max(H - L, H - C_{t-1} , L - C_{t-1})$
20	atr_norm	$\text{ATR}(14)/C_t$
21	vol_5d	$\text{std}_5(\text{ret_1d})\sqrt{252}$
22	vol_10d	$\text{std}_{10}(\text{ret_1d})\sqrt{252}$
23	vol_20d	$\text{std}_{20}(\text{ret_1d})\sqrt{252}$
24	vol_60d	$\text{std}_{60}(\text{ret_1d})\sqrt{252}$
25	vol_ratio_5_20	$\text{vol_5d} / \text{vol_20d}$
26	vol_ratio_20_60	$\text{vol_20d} / \text{vol_60d}$

#	Column	Definition
<i>Oscillators (9)</i>		
27	rsi_14	Wilder RSI(14): $100 - 100/(1 + \overline{\text{gain}}/\overline{\text{loss}})$, exponential smoothing $\alpha = 1/14$
28	rsi_7	Wilder RSI(7), $\alpha = 1/7$
29	macd	$(\text{EMA}_{12}(C) - \text{EMA}_{26}(C))/C_t$
30	macd_sig	$\text{EMA}_9(\text{MACD line})/C_t$
31	macd_hist	$(\text{MACD line} - \text{signal line})/C_t$
32	stoch_k	$\%K = 100 (C - \min_{14} L)/(\max_{14} H - \min_{14} L)$
33	stoch_d	$\%D$: 3-day mean of $\%K$
34	stoch_diff	$\%K - \%D$
35	williams_r	$-100 (\max_{14} H - C)/(\max_{14} H - \min_{14} L)$
<i>Volume and flow (8)</i>		
36	vol_log	$\ln(V + 1)$
37	vol_ratio	$V_t/\text{MA}_{20}(V)$
38	obv_ratio	$\text{OBV}_t/ \text{MA}_{20}(\text{OBV}) - 1$, $\text{OBV}_t = \sum \text{sign}(\Delta C) V$ from the first downloaded day
39	obv_pendiente	$((\text{OBV}_t - \text{OBV}_{t-5})/\text{OBV}_{t-5})/C_t$ (OBV slope)
40	vwap_dist	$(C - \text{VWAP}_{20})/\text{VWAP}_{20}$, $\text{VWAP}_{20} = \sum_{20} \text{TP} V / \sum_{20} V$, $\text{TP} = (H + L + C)/3$
41	cmf_20	Chaikin money flow $\sum_{20} \text{CLV} V / \sum_{20} V$, $\text{CLV} = ((C - L) - (H - C))/(H - L)$
42	mfi_14	Money flow index over 14 days on $\text{TP} V$
43	vol_trend	$\text{MA}_{20}(V)/\text{MA}_{60}(V) - 1$
<i>Candle patterns (7)</i>		
44	rango_rel	$(H - L)/C$ (relative range)
45	cambio_intra	$(C - O)/O$ (intraday change)
46	cuerpo_rel	$ C - O /(H - L)$ (body / range)
47	sombra_sup	$(H - \max(O, C))/\text{ATR}(14)$ (upper shadow)
48	sombra_inf	$(\min(O, C) - L)/\text{ATR}(14)$ (lower shadow)
49	gap_apertura	$(O_t - C_{t-1})/C_{t-1}$ (opening gap)
50	hl_ratio	$(C - L)/(H - L)$ (position of the close in the range)
<i>Seasonality (5)</i>		
51	dia_sin	$\sin(2\pi d/5)$, $d = \text{weekday}$ (Monday = 0)
52	dia_cos	$\cos(2\pi d/5)$
53	mes_sin	$\sin(2\pi m/12)$, $m = \text{month}$ (1–12)
54	mes_cos	$\cos(2\pi m/12)$
55	semana_mes	$(w - 2.5)/2.5$, $w = \lfloor (\text{day of month} - 1)/7 \rfloor + 1$
<i>Market context (6)</i>		
56	SP500_ret	$\ln(\text{SPY}_t/\text{SPY}_{t-1})$
57	SP500_vol20	$\text{std}_{20}(\text{SP500_ret})\sqrt{252}$
58	SP500_mom20	$\text{SPY}_t/\text{SPY}_{t-19} - 1$
59	VIX	close of the CBOE VIX index ($\sim \text{VIX}$)
60	VIX_change	$\text{VIX}_t/\text{VIX}_{t-1} - 1$
61	VIX_norm	$(\text{VIX} - \text{MA}_{252}(\text{VIX}))/\text{std}_{252}(\text{VIX})$

S2 Per-class F1 for every run (paper §5.1)

F1-macro and per-class F1 on the 2025 test set for the 120 runs (5 models $\times$ 3 experiments $\times$ (1 global + 7 per-ticker)). “Global” models are trained on the seven tickers together and scored on all of them; per-ticker models are trained and scored on one ticker. Look-back is the number of past days given to the sequence models.

Table S2: Test F1-macro and per-class F1 of the 120 runs (test year 2025).

Model	Training	Exp	Look-back	F1-macro	F1 SELL	F1 HOLD	F1 BUY
LR	Global	A	–	0.411	0.357	0.521	0.354
LR	Global	B	–	0.385	0.364	0.512	0.278
LR	Global	C	–	0.401	0.355	0.533	0.314
LR	AAPL	A	–	0.394	0.352	0.514	0.315
LR	AAPL	B	–	0.384	0.333	0.528	0.290
LR	AAPL	C	–	0.369	0.306	0.483	0.317
LR	NVDA	A	–	0.418	0.410	0.549	0.296
LR	NVDA	B	–	0.421	0.405	0.598	0.261
LR	NVDA	C	–	0.299	0.418	0.479	0.000
LR	TSLA	A	–	0.408	0.428	0.492	0.303
LR	TSLA	B	–	0.360	0.406	0.475	0.198
LR	TSLA	C	–	0.359	0.396	0.422	0.258
LR	AMZN	A	–	0.423	0.410	0.500	0.358
LR	AMZN	B	–	0.430	0.469	0.523	0.297
LR	AMZN	C	–	0.386	0.473	0.447	0.237
LR	MSFT	A	–	0.408	0.315	0.568	0.340
LR	MSFT	B	–	0.376	0.288	0.554	0.286
LR	MSFT	C	–	0.382	0.339	0.562	0.246
LR	GOOGL	A	–	0.420	0.279	0.602	0.378
LR	GOOGL	B	–	0.384	0.387	0.461	0.302
LR	GOOGL	C	–	0.358	0.408	0.396	0.271
LR	META	A	–	0.410	0.353	0.462	0.414
LR	META	B	–	0.387	0.254	0.548	0.358
LR	META	C	–	0.357	0.194	0.495	0.383
XGBoost	Global	A	–	0.396	0.357	0.486	0.346
XGBoost	Global	B	–	0.386	0.367	0.453	0.338
XGBoost	Global	C	–	0.373	0.353	0.426	0.341
XGBoost	AAPL	A	–	0.384	0.336	0.529	0.286
XGBoost	AAPL	B	–	0.336	0.325	0.426	0.257
XGBoost	AAPL	C	–	0.339	0.306	0.434	0.275
XGBoost	NVDA	A	–	0.380	0.362	0.474	0.304
XGBoost	NVDA	B	–	0.338	0.324	0.400	0.291
XGBoost	NVDA	C	–	0.372	0.392	0.482	0.241
XGBoost	TSLA	A	–	0.406	0.340	0.497	0.382
XGBoost	TSLA	B	–	0.441	0.385	0.505	0.435
XGBoost	TSLA	C	–	0.395	0.317	0.461	0.408
XGBoost	AMZN	A	–	0.352	0.309	0.374	0.374
XGBoost	AMZN	B	–	0.390	0.363	0.419	0.389
XGBoost	AMZN	C	–	0.377	0.367	0.404	0.361
XGBoost	MSFT	A	–	0.425	0.381	0.492	0.403
XGBoost	MSFT	B	–	0.399	0.387	0.436	0.372
XGBoost	MSFT	C	–	0.331	0.327	0.419	0.248
XGBoost	GOOGL	A	–	0.344	0.258	0.419	0.354
XGBoost	GOOGL	B	–	0.336	0.286	0.394	0.329
XGBoost	GOOGL	C	–	0.328	0.328	0.385	0.271
XGBoost	META	A	–	0.342	0.323	0.366	0.336
XGBoost	META	B	–	0.347	0.271	0.400	0.370
XGBoost	META	C	–	0.321	0.290	0.314	0.359
LSTM	Global	A	60	0.389	0.371	0.522	0.274
LSTM	Global	B	60	0.370	0.286	0.522	0.301
LSTM	Global	C	20	0.370	0.281	0.473	0.356
LSTM	AAPL	A	60	0.358	0.121	0.528	0.424
LSTM	AAPL	B	20	0.383	0.286	0.557	0.306
LSTM	AAPL	C	20	0.322	0.078	0.459	0.429
LSTM	NVDA	A	20	0.154	0.375	0.000	0.087
LSTM	NVDA	B	20	0.328	0.250	0.635	0.099

Model	Training	Exp	Look-back	F1-macro	F1 SELL	F1 HOLD	F1 BUY
LSTM	NVDA	C	20	0.277	0.370	0.462	0.000
LSTM	TSLA	A	20	0.202	0.057	0.111	0.439
LSTM	TSLA	B	60	0.291	0.267	0.254	0.353
LSTM	TSLA	C	20	0.299	0.057	0.465	0.374
LSTM	AMZN	A	20	0.351	0.209	0.438	0.405
LSTM	AMZN	B	20	0.335	0.216	0.386	0.404
LSTM	AMZN	C	20	0.363	0.216	0.533	0.341
LSTM	MSFT	A	20	0.392	0.447	0.455	0.273
LSTM	MSFT	B	20	0.353	0.436	0.483	0.139
LSTM	MSFT	C	60	0.332	0.346	0.547	0.102
LSTM	GOOGL	A	20	0.380	0.265	0.503	0.372
LSTM	GOOGL	B	60	0.415	0.234	0.571	0.439
LSTM	GOOGL	C	20	0.377	0.115	0.600	0.416
LSTM	META	A	20	0.359	0.323	0.432	0.324
LSTM	META	B	60	0.364	0.336	0.420	0.337
LSTM	META	C	60	0.310	0.383	0.252	0.295
CNN-LSTM	Global	A	60	0.376	0.277	0.492	0.360
CNN-LSTM	Global	B	20	0.384	0.273	0.546	0.331
CNN-LSTM	Global	C	60	0.371	0.266	0.525	0.323
CNN-LSTM	AAPL	A	20	0.353	0.284	0.487	0.288
CNN-LSTM	AAPL	B	60	0.325	0.215	0.494	0.266
CNN-LSTM	AAPL	C	20	0.307	0.052	0.529	0.339
CNN-LSTM	NVDA	A	20	0.302	0.365	0.542	0.000
CNN-LSTM	NVDA	B	20	0.311	0.300	0.633	0.000
CNN-LSTM	NVDA	C	20	0.339	0.410	0.546	0.060
CNN-LSTM	TSLA	A	20	0.413	0.277	0.535	0.427
CNN-LSTM	TSLA	B	20	0.352	0.426	0.395	0.235
CNN-LSTM	TSLA	C	20	0.383	0.357	0.469	0.321
CNN-LSTM	AMZN	A	60	0.384	0.200	0.569	0.384
CNN-LSTM	AMZN	B	20	0.418	0.375	0.523	0.357
CNN-LSTM	AMZN	C	20	0.378	0.286	0.472	0.376
CNN-LSTM	MSFT	A	20	0.370	0.272	0.526	0.311
CNN-LSTM	MSFT	B	20	0.329	0.368	0.562	0.058
CNN-LSTM	MSFT	C	20	0.380	0.305	0.510	0.325
CNN-LSTM	GOOGL	A	60	0.391	0.275	0.571	0.326
CNN-LSTM	GOOGL	B	20	0.315	0.000	0.613	0.333
CNN-LSTM	GOOGL	C	20	0.343	0.385	0.566	0.079
CNN-LSTM	META	A	20	0.361	0.348	0.546	0.188
CNN-LSTM	META	B	20	0.346	0.356	0.476	0.207
CNN-LSTM	META	C	20	0.360	0.398	0.524	0.160
CNN 1D	Global	A	60	0.361	0.221	0.532	0.329
CNN 1D	Global	B	20	0.343	0.113	0.515	0.402
CNN 1D	Global	C	20	0.373	0.288	0.507	0.323
CNN 1D	AAPL	A	20	0.295	0.305	0.232	0.347
CNN 1D	AAPL	B	20	0.363	0.226	0.545	0.318
CNN 1D	AAPL	C	60	0.159	0.000	0.113	0.363
CNN 1D	NVDA	A	60	0.409	0.378	0.503	0.345
CNN 1D	NVDA	B	20	0.316	0.000	0.576	0.373
CNN 1D	NVDA	C	20	0.286	0.404	0.454	0.000
CNN 1D	TSLA	A	20	0.408	0.403	0.436	0.385
CNN 1D	TSLA	B	20	0.406	0.270	0.585	0.362
CNN 1D	TSLA	C	60	0.314	0.236	0.364	0.343
CNN 1D	AMZN	A	20	0.371	0.209	0.507	0.397
CNN 1D	AMZN	B	20	0.325	0.093	0.534	0.348
CNN 1D	AMZN	C	20	0.282	0.000	0.571	0.275
CNN 1D	MSFT	A	20	0.387	0.260	0.516	0.386
CNN 1D	MSFT	B	60	0.279	0.000	0.564	0.274

Model	Training	Exp	Look-back	F1-macro	F1 SELL	F1 HOLD	F1 BUY
CNN 1D	MSFT	C	20	0.286	0.049	0.523	0.286
CNN 1D	GOOGL	A	20	0.379	0.127	0.535	0.474
CNN 1D	GOOGL	B	60	0.328	0.387	0.598	0.000
CNN 1D	GOOGL	C	60	0.336	0.091	0.515	0.403
CNN 1D	META	A	60	0.396	0.420	0.443	0.325
CNN 1D	META	B	20	0.304	0.000	0.556	0.357
CNN 1D	META	C	20	0.297	0.355	0.424	0.111

S3 Economic metrics for Exps A and C (paper §7.1)

Same metrics as Table 5 of the paper (which reports Exp B) for the global models of the other two experiments: long on BUY, short on SELL, flat on HOLD, one-day holding period, no transaction costs; Sharpe is annualized. Best value per column in bold.

Table S3: Economic metrics, Exp A (10 training years) GLOBAL, test = 2025.

Model	F1	Win rate	Profit factor	Max drawdown	Sharpe
LR	0.411	0.528	1.296	−0.551	+1.067
XGBoost	0.396	0.505	1.143	−0.464	+0.575
LSTM	0.389	0.510	1.284	−0.392	+0.965
CNN-LSTM	0.376	0.502	1.187	−0.460	+0.686
CNN 1D	0.361	0.467	1.007	−0.365	+0.025

Table S4: Economic metrics, Exp C (4 training years) GLOBAL, test = 2025.

Model	F1	Win rate	Profit factor	Max drawdown	Sharpe
LR	0.401	0.513	1.297	−0.445	+1.023
XGBoost	0.373	0.500	1.090	−0.428	+0.377
LSTM	0.370	0.496	1.039	−0.631	+0.163
CNN-LSTM	0.371	0.508	1.182	−0.463	+0.624
CNN 1D	0.373	0.494	0.924	−0.666	−0.317

S4 Sharpe ratio by model and ticker (paper §7.3)

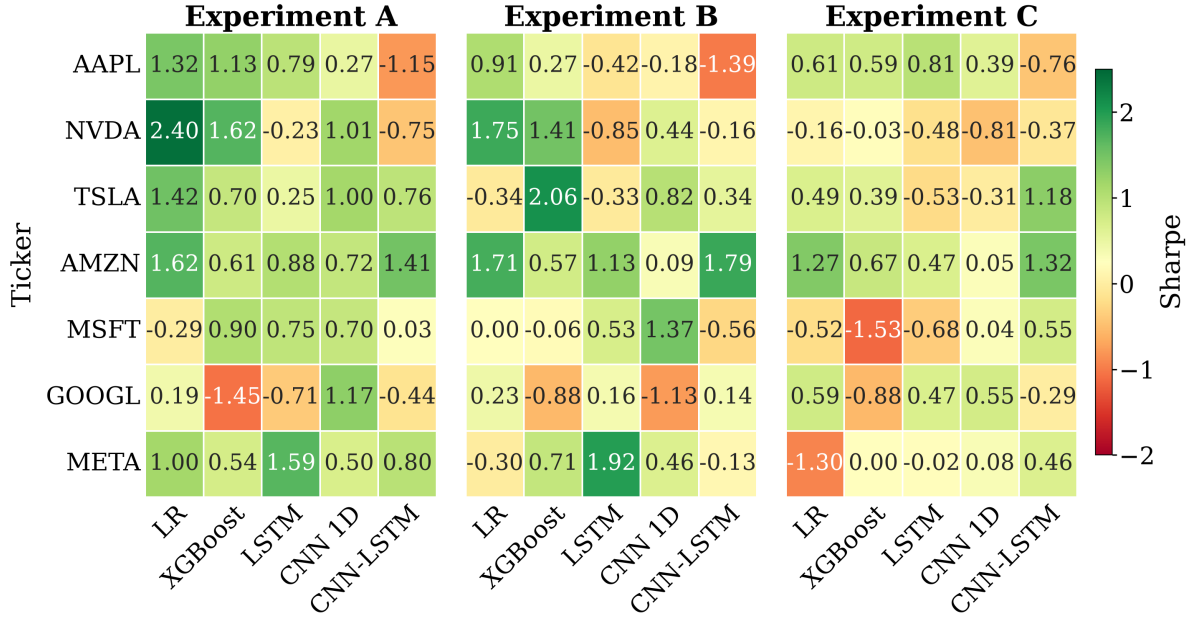


Figure S1: Annualized Sharpe ratio of the per-ticker models, per model and ticker, for the three experiments (test = 2025). These are the 105 values whose variance decomposition is given in Section 7.3 of the paper.

S5 Code, data manifest, random seeds and library versions (paper §11)

Code. The complete pipeline is publicly available at <https://github.com/barudust/TT>; a snapshot of the repository for this paper is tagged micai2026 (<https://github.com/barudust/TT/tree/micai2026>). Table S5 lists where each part of the paper is implemented.

Table S5: Where each part of the paper is implemented (paths relative to the repository root).

Part of the paper	Files
§3 Data download, features and label	scripts_v1/01_build_raw_dataset.py
§3.4 Temporal splits	scripts_opt/common_v4.py
§6 Statistical feature pre-selection	scripts_opt/analysis_feature_selection.py
§7 Benchmark of 120 runs	scripts_opt/train_all_v4.py scripts_opt/consolidar_v4.py RESULTADOS_OPTIMIZADOS/v4/resultados_v4.csv (results)
§8 Robustness study	scripts_opt/run_v5.py scripts_opt/common_v5.py scripts_opt/reporte_v5.py RESULTADOS_OPTIMIZADOS/v5/ (results)
Figures	scripts_opt/plots_paper.py
Reproduction checks	RESULTADOS_OPTIMIZADOS/v5/g0_replicacion.csv (results) scripts_opt/replicar_lr_v4.py RESULTADOS_OPTIMIZADOS/v4/replica_lr_v4.csv (results)
This document	paper_review/supplementary/build_supplementary.py

Data manifest. Source: Yahoo Finance through the `yfinance` library, daily bars adjusted for splits and dividends (`auto_adjust=True`), downloaded for 2013-01-01 to 2025-12-31; the first year serves as warm-up for

the indicators. Market context: SPY and $\hat{VIX}$ over the same window. Label: one-day forward log-return; BUY if it reaches the 70th percentile and SELL if it falls to the 30th percentile of the previous 252 trading days of the same ticker (thresholds shifted by one day, so no future information enters them), HOLD otherwise. Rows per ticker: training 2,516 (Exp A, 2014–2023), 1,509 (Exp B, 2018–2023) and 1,006 (Exp C, 2020–2023); validation 252 (2024); test 248 (2025). Global models use the seven tickers together (1,736 test rows); in Section 7 the sequence models were scored on the test days with a complete look-back window (1,596 rows at look-back 20 and 1,316 at 60), which the protocol of Section 8 aligns to the same 1,736 rows. Table S6 lists the stored datasets.

Table S6: Stored datasets `tesis_ml_stocks/01_raw_datasets/<TICKER>_raw.parquet` (one row per trading day: the 61 features, the raw OHLCV, and the label with its forward return and percentile thresholds) and their SHA-256 checksums.

Ticker	First day	Last day	Rows	SHA-256
AAPL	2013-12-31	2025-12-29	3,017	fe020d29efa861c96dad12a5d7e2070bcc876e5db3146ec6b8083ec69df3d2eb
NVDA	2013-12-31	2025-12-29	3,017	6e5afa41aacb7556402f391bd07460f1aaba5695f5cada58042278766ac86b0c
TSLA	2013-12-31	2025-12-29	3,017	83535218b012e791db6fa6b60a8f4459b0ee6a14541c53203b74edd03fc7d6e0
AMZN	2013-12-31	2025-12-29	3,017	70518ea782bd65bb7b5fb10147c026a4c8bd7ff83255f51baa0565764d386300
MSFT	2013-12-31	2025-12-29	3,017	22bd02e399884a6e06399def52f2cef2e2d1cc5ab9e6402d288a0ce2dbaa5db4
GOOGL	2013-12-31	2025-12-29	3,017	fc19f91c45c0c524ebaaa93be82f28a31befd62a8fbf8dc89a2cea256eebba5c
META	2013-12-31	2025-12-29	3,017	9fc4d44cc1913040225335329b332102f19bb3ddd97a7646fdf0b68318e69780

Random seeds. Benchmark (§7): LR is trained with seeds {42, 1, 7, 2024, 100} and the final class is the majority vote of the five models; XGBoost, LSTM, CNN 1D and CNN-LSTM are trained with seeds {42, 1, 7} and their class probabilities are averaged; the Optuna search of XGBoost uses a TPE sampler with seed 42. Robustness study (§8): Optuna TPE sampler with seed 42 (25 random start-up trials) and a median pruner for the deep models; every frozen configuration is evaluated on the test year with seeds {42, 1, 7, 2024, 100}.

Library versions and hardware. PyTorch 2.12 (CUDA 12.8), scikit-learn 1.7.2, XGBoost 3.2.0 and Optuna 4.8, on an NVIDIA GeForce RTX 5060 Ti. These versions were recorded for the robustness study (§8). Two checks show that this environment reproduces the benchmark of Section 7. The 18 global sequence-model runs (LSTM, CNN 1D and CNN-LSTM; three experiments; look-backs 20 and 60), re-run with the code of Section 8 set to the protocol of Section 7, give the same test F1-macro as the published runs, to the four decimals stored, in 18 of 18 cases. The 24 LR runs, re-run with the original code (Python 3.13.5, scikit-learn 1.7.2), give identical F1-macro, per-class F1, Sharpe, win rate, profit factor and maximum drawdown in 24 of 24 cases (Table S5, “Reproduction checks”). The XGBoost runs and the per-ticker sequence-model runs were not re-run.